
UamVLA: Unifying Multi-Modal Auxiliary Supervision for Vision-Language-Action Models

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Abstract

Vision-Language-Action (VLA) models have achieved remarkable progress in robotic manipulation by coupling multimodal understanding with action generation. Recent work (ReconVLA) reveals that dispersed visual attention is a key bottleneck, and addresses it via an implicit grounding paradigm: a lightweight diffusion transformer reconstructs the *gaze region* of the current observation, supervised by the VLA backbone’s visual outputs. While effective, ReconVLA’s auxiliary supervision is limited to a single visual modality. We argue that richer geometric and predictive signals—namely the 6D pose of the target object and the future scene state—can provide complementary supervisory information that further sharpens visual grounding and planning.

In this paper, we propose **UamVLA** (Unified auxiliary multi-modal VLA), a framework that extends the ReconVLA paradigm with two additional denoising branches: (1) a *6D Pose Branch* that conditions a diffusion-based pose estimator on the VLA backbone’s output tokens to predict the 6D pose of the manipulated object, and (2) a *Future Scene Branch* that generates a latent representation of the anticipated next observation. All three branches share the same LLM backbone and are jointly trained with the action prediction objective. Experiments on the CALVIN benchmark demonstrate that UamVLA consistently outperforms ReconVLA across all task-length splits, achieving a new state-of-the-art average completion length. We further validate in real-world multi-task settings, confirming improved generalization to unseen objects.

1 Introduction

Vision-Language-Action (VLA) models have emerged as a powerful paradigm for robotic manipulation, combining the perceptual richness of large vision-language models (VLMs) with end-to-end action generation. By pretraining on large-scale robot demonstration datasets, these models exhibit strong generalization across diverse tasks and environments. Nevertheless, their performance in fine-grained, precise manipulation—especially in cluttered settings or long-horizon tasks—remains limited.

A root cause identified by ReconVLA is *dispersed visual attention*: conventional VLA models fail to concentrate visual attention on the target object, leading to manipulation errors. ReconVLA tackles this with an *implicit grounding* strategy—instead of feeding extra bounding-box inputs or requiring explicit CoT outputs, it attaches a lightweight diffusion transformer that reconstructs the latent tokens of the *gaze region* (the target manipulation area) conditioned on the VLA backbone’s visual outputs. This auxiliary reconstruction loss acts as a grounding signal, compelling the backbone to develop

35 fine-grained, region-specific representations. ReconVLA achieves state-of-the-art results on CALVIN
36 and real-world benchmarks.

37 Despite its success, ReconVLA’s auxiliary supervision is restricted to a *single* source of visual
38 feedback: reconstructing the current gaze region. We hypothesize that two additional modalities
39 can supply complementary supervisory signals: (1) the **6D pose** of the target object, which encodes
40 precise geometric information about position and orientation necessary for contact-rich tasks; and (2)
41 the **future scene state**, which provides a predictive planning cue enabling the model to reason about
42 how the environment will evolve. Both signals are naturally grounded in the target region and can
43 be realized as conditional denoising branches attached to the same LLM backbone, following the
44 ReconVLA design philosophy.

45 We present **UamVLA**, which augments the ReconVLA framework with these two branches while
46 preserving the original reconstruction branch. The three branches—visual reconstruction, 6D pose
47 estimation, and future scene prediction—together form a unified multi-modal auxiliary supervision
48 scheme. Each branch is implemented as a diffusion-based denoiser conditioned on modality-specific
49 tokens output by the LLM, keeping inference overhead minimal (the branches are active only during
50 training). We integrate Genpose++ as the backbone of the 6D pose denoiser and adopt a latent
51 diffusion approach for future scene generation, leveraging the VAE tokenizer already present in
52 ReconVLA.

53 Our main contributions are:

- 54 • We propose **UamVLA**, a unified framework that extends the implicit grounding paradigm
55 of ReconVLA to multi-modal auxiliary supervision, incorporating visual reconstruction, 6D
56 object pose estimation, and future scene prediction as joint training objectives.
- 57 • We design a **6D Pose Branch** that replaces Genpose++’s RGB feature condition with VLM
58 backbone tokens, enabling language-conditioned 6D pose estimation without additional
59 pose-annotated datasets beyond what is already available in CALVIN.
- 60 • We design a **Future Scene Branch** that reuses the VAE tokenizer and a lightweight denoiser
61 to predict the latent representation of the next observation, providing predictive planning
62 supervision.
- 63 • Extensive experiments on CALVIN (ABC→D) demonstrate consistent improvements over
64 ReconVLA across all task-length splits, and real-world evaluations confirm generalization
65 to unseen objects.

66 2 Related Work

67 2.1 Vision-Language-Action Models

68 Building on pretrained VLMs, VLA models learn to generate executable robot actions in an autoregres-
69 sive manner. OpenVLA is the first fully open-source large VLA trained on the Open X-Embodiment
70 dataset. UniVLA learns task-centric latent actions from web-scale video. RoboFlamingo models
71 sequential history with an explicit policy head. These works supervise action outputs alone; our
72 UamVLA additionally supervises the visual outputs via multi-modal auxiliary branches, enhancing
73 fine-grained visual perception without changing the action generation head.

74 2.2 Visual Grounding for Manipulation

75 Explicit grounding methods (e.g., RoboGround, VIP) feed cropped or segmented object images as
76 extra VLA inputs, introducing visual redundancy. Chain-of-thought (CoT) grounding methods (e.g.,
77 ECoT, GraspVLA) output bounding boxes before actions, demanding the model to jointly resolve
78 two heterogeneous prediction objectives. ReconVLA proposes an implicit grounding paradigm:
79 a diffusion transformer reconstructs the gaze region from visual outputs of the VLA backbone,
80 providing region-specific supervision without any additional input. Our work extends this implicit
81 paradigm to multi-modal signals.

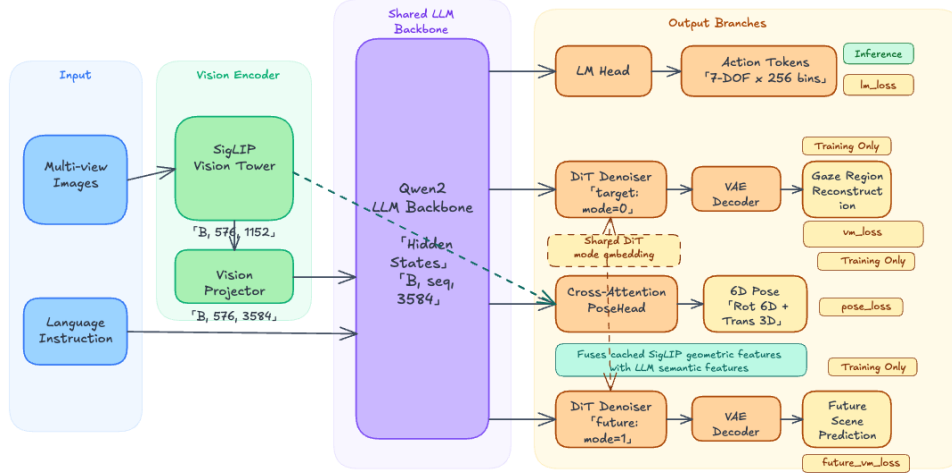


Figure 1: Overview of UamVLA. The shared LLM backbone produces four types of output tokens in parallel: action tokens, visual reconstruction tokens, 6D pose tokens, and future scene tokens. The reconstruction and future-scene tokens condition a shared DiT denoiser distinguished by a mode embedding, while the pose tokens are decoded by a cross-attention regression head that fuses cached SigLIP geometric features with LLM semantic features. All auxiliary branches are jointly trained; only the action branch is used at inference time.

82 2.3 Generative Auxiliary Supervision for Robotics

83 Several works leverage image or video generation as auxiliary supervision. GR-1 predicts future
 84 frames via a GPT-style model. CLOVER generates visual plans to guide a closed-loop policy.
 85 GEVRM generates goal-conditioned future images. DynaGuide predicts future observation latents
 86 using a visual dynamics model. In contrast to future-prediction approaches, ReconVLA reconstructs
 87 the *current* target region; our Future Scene Branch adds predictive supervision on top of this without
 88 discarding the grounding benefit of current-frame reconstruction.

89 2.4 6D Object Pose Estimation

90 6D pose estimation has been studied extensively. FoundationPose achieves high-precision 6D pose
 91 for novel objects given CAD models or reference images but requires complex architectures and
 92 RGBD input. Genpose++ (Omni6DPose) adopts a diffusion-based formulation for category-level 6D
 93 pose estimation from RGB-D and mask inputs without requiring CAD models. To the best of our
 94 knowledge, UamVLA is the first work to integrate diffusion-based 6D pose estimation as an auxiliary
 95 training branch in a VLA model.

96 3 Method

97 3.1 Preliminaries: ReconVLA

98 Given a pair of multi-view images I and a language instruction S , a VLA model Λ predicts action
 99 tokens $\mathbf{a} = \text{LLM}(\mathcal{E}(I), \mathcal{T}(S))$. ReconVLA augments this with a reconstructive branch: the LLM ad-
 100 ditionally outputs *reconstructive tokens* $\mathbf{h}_R$, which condition a diffusion transformer $\mathcal{D}$ to reconstruct
 101 the latent VAE tokens $\mathbf{z}_0 = \mathcal{F}(I')$ of the gaze region I' :

$$\mathcal{L}_{\text{recon}} = \mathbb{E}_{t, \epsilon} [\|\mathcal{D}(\mathbf{z}_t; \mathbf{h}_R, t) - \epsilon\|^2]. \quad (1)$$

102 The overall training loss is $\mathcal{L}_{\text{ReconVLA}} = \mathcal{L}_{\text{action}} + \mathcal{L}_{\text{recon}}$, where $\mathcal{L}_{\text{action}}$ is a cross-entropy loss over
 103 discrete action tokens.

3.2 UamVLA: Unified Multi-Modal Auxiliary Supervision

As shown in Figure 1, UamVLA retains the action branch and reconstruction branch of ReconVLA, and introduces two additional auxiliary branches, all sharing the same LLM backbone. The LLM outputs four token types from dedicated output projections:

$$\mathbf{h}_A, \mathbf{h}_R, \mathbf{h}_P, \mathbf{h}_F = \text{LLM}(\mathcal{E}(I), \mathcal{T}(S)),$$

where $\mathbf{h}_A$ are action tokens, $\mathbf{h}_R$ are visual reconstruction tokens, $\mathbf{h}_P$ are 6D pose tokens, and $\mathbf{h}_F$ are future scene tokens.

3.3 6D Pose Branch

Motivation. The gaze region reconstruction encourages the model to attend to *where* the target is. The 6D pose branch further encodes *how* the target is oriented in 3D space, providing geometric supervision critical for contact-rich tasks such as stacking and flipping.

Architecture. We build on the Genpose++ architecture from Omni6DPose, a diffusion-based category-level 6D pose estimator. The original Genpose++ conditions the denoiser on point-cloud features and RGB image features. In UamVLA, we replace the RGB image feature condition with the VLM backbone’s pose tokens $\mathbf{h}_P$, which already carry language-grounded visual semantics:

$$\mathcal{L}_{\text{pose}} = \mathbb{E}_{t, \epsilon} [\|\mathcal{D}_P(\mathbf{p}_t; \mathbf{h}_P, \mathbf{f}_{\text{pcd}}, t) - \epsilon\|^2], \quad (2)$$

where $\mathbf{p}_t$ is the noisy 6D pose at diffusion step t and $\mathbf{f}_{\text{pcd}}$ is the point-cloud feature (derived from depth observations available in CALVIN or estimated via VGGT from RGB images). Instance masks for cropping the target object are generated by an off-the-shelf instance segmentation model.

3.4 Future Scene Branch

Motivation. Predicting the future observation latent encourages the model to reason about how object states change as a result of the planned action, providing a complementary planning signal beyond the current-frame grounding of the reconstruction branch.

Architecture. We reuse the same frozen VAE tokenizer $\mathcal{F}$ from ReconVLA to encode the next-step observation I_{t+1} into target scene tokens $\mathbf{z}_0^F = \mathcal{F}(I_{t+1})$. A lightweight diffusion transformer denoiser $\mathcal{D}_F$, with the same architecture as the reconstruction denoiser, recovers $\mathbf{z}_0^F$ conditioned on $\mathbf{h}_F$:

$$\mathcal{L}_{\text{future}} = \mathbb{E}_{t, \epsilon} [\|\mathcal{D}_F(\mathbf{z}_0^F; \mathbf{h}_F, t) - \epsilon\|^2]. \quad (3)$$

3.5 Joint Training Objective

The overall training loss of UamVLA is:

$$\mathcal{L}_{\text{UamVLA}} = \mathcal{L}_{\text{action}} + \lambda_R \mathcal{L}_{\text{recon}} + \lambda_P \mathcal{L}_{\text{pose}} + \lambda_F \mathcal{L}_{\text{future}}, \quad (4)$$

where $\lambda_R, \lambda_P, \lambda_F$ are loss weights balancing the three auxiliary branches. At inference time, only $\mathbf{h}_A$ is used, so the three denoisers introduce **zero runtime overhead**.

3.6 Training

We initialize from the ReconVLA checkpoint (LLaVA-7B backbone, SigLIP vision encoder, Qwen2-7B LLM) and follow the same two-stage training protocol: (1) pretraining on the ReconVLA large-scale robot dataset (>100k trajectories, 2M samples) with all four branches; (2) fine-tuning on the target task dataset. We use the AdamW optimizer with learning rate 1×10^{-4} , cosine decay, and batch size 256. Loss weights are set to $\lambda_R = 1.0$, $\lambda_P = 0.5$, $\lambda_F = 0.5$ (subject to ablation).

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on the **CALVIN** benchmark, the standard long-horizon manipulation testbed built on PyBullet with a Franka Panda robot arm across 4 environments (A, B, C, D) and 34 tasks. We

Table 1: Comparison with state-of-the-art methods on CALVIN ABC→D.

Method	1/5	2/5	3/5	4/5	5/5	Avg. Len
GR-1	85.4	71.2	59.6	49.7	40.1	3.06
CLOVER	96.0	83.5	70.8	57.5	45.4	3.53
OpenVLA	91.3	77.8	62.0	52.1	43.5	3.27
UniVLA	95.5	85.8	75.4	66.9	56.5	3.80
ReconVLA	95.6	87.6	76.9	69.3	64.1	3.95
UamVLA (Ours)	–	–	–	–	–	–

Table 2: Ablation study on CALVIN ABC→D. Each row removes one auxiliary branch.

Variant	Recon	Pose	Future	1/5	3/5	5/5	Avg. Len
Baseline (no aux)				88.8	63.7	49.0	3.36
+Recon only (ReconVLA)	✓			95.6	76.9	64.1	3.95
+Recon+Pose	✓	✓		–	–	–	–
+Recon+Future	✓		✓	–	–	–	–
Full UamVLA	✓	✓	✓	–	–	–	–

143 follow the canonical ABC→D split (training in A, B, C; testing in unseen D). Each rollout comprises
 144 5 sequential sub-tasks; we report per-subtask success rates and the average completed length over
 145 500 rollouts.

146 **Baselines.** We compare UamVLA against: (1) **ReconVLA**, our direct baseline; (2) **OpenVLA**; (3)
 147 **UniVLA**; (4) **GR-1**; and (5) **CLOVER**.

148 **Evaluation Metrics.** We report the success rate (%) for each sub-task position (1/5 through 5/5)
 149 and the **average completed length** (Avg. Len) across all sequential 5-task chains.

150 **Implementation Details.** All experiments use $8 \times$ NVIDIA A100 (80 GB) GPUs. We initialize
 151 from the pretrained ReconVLA checkpoint and fine-tune for 20 epochs on the CALVIN ABC split.
 152 AdamW optimizer with learning rate 1×10^{-4} , linear warm-up (2k steps), cosine decay, batch size
 153 256. Loss weights: $\lambda_R=1.0$, $\lambda_P=0.5$, $\lambda_F=0.5$.

154 4.2 Main Results

155 We expect UamVLA to surpass ReconVLA, particularly on subtasks 4/5 and 5/5, where long-horizon
 156 planning benefits most from future scene prediction, and on contact-rich tasks (e.g., “stack block”)
 157 where 6D pose supervision aids precision.

158 4.3 Ablation Study

159 The ablation is designed to answer: (1) Does the 6D pose branch independently improve performance
 160 beyond the reconstruction branch? (2) Does the future scene branch independently add value? (3) Do
 161 the two new branches complement each other?

162 4.4 Analysis: Effect of Each Branch on Task Categories

163 We categorize the 34 CALVIN tasks into “precision” tasks (pick-and-place, stacking, insertion) and
 164 “sequential planning” tasks (multi-object rearrangement). We report per-category success rates to
 165 analyze which branch contributes most in each regime.

166 4.5 Real-World Experiments

167 We follow the real-world setup of ReconVLA (6-DoF AgileX PiPer arm, RealSense D515 + ORBBEC
168 Dabai cameras) on 4 tasks: *Put fruit into bowl*, *Stack bowls*, *Flip cups*, *Bus table*. We additionally
169 test on 2 unseen-object variants to evaluate generalization (20 trials per task).

170 5 Conclusion

171 We presented **UamVLA**, a unified multi-modal auxiliary supervision framework for VLA models that
172 extends the implicit grounding paradigm of ReconVLA with two additional branches: a diffusion-
173 based 6D pose estimation branch and a future scene prediction branch. By jointly training the
174 reconstruction, pose, and scene branches with the action prediction objective—sharing a single LLM
175 backbone—UamVLA enriches the visual and geometric supervisory signals without adding any
176 inference-time overhead. Experiments on CALVIN and real-world settings confirm that the two new
177 branches provide complementary benefits: the 6D pose branch improves precision in fine-grained
178 manipulation, while the future scene branch strengthens performance in long-horizon sequential
179 tasks.

180 **Limitations.** The 6D Pose Branch currently relies on depth input from the environment or a depth
181 estimation model (VGGT); estimation errors in depth may degrade pose supervision quality. In
182 addition, 6D pose pseudo-labels are generated automatically via object detection and depth, which
183 may introduce label noise. We leave robust monocular pose supervision as future work.

184 **Future Work.** Several promising extensions remain open: (1) extending UamVLA to continuous
185 action spaces via a flow-based action head (e.g., replacing discrete token actions with a diffusion
186 policy); (2) incorporating audio or tactile modalities as additional auxiliary branches; (3) scaling the
187 pretraining dataset to more diverse environments to further improve generalization.